🔍 What is Wireshark?

Wireshark is a free and open-source **network protocol analyzer**. It captures and displays data packets **in real-time**, enabling users to:

- Analyze network traffic
- Troubleshoot network issues
- Detect security anomalies
- Learn network protocols in depth

It supports **hundreds of protocols** like TCP, UDP, HTTP, DNS, ICMP, and SSL/TLS.

🔍 1. Creating Filters in Wireshark

Filters in Wireshark help **narrow down traffic** to only what's relevant. There are two major types:

Capture Filters

- Applied **before** capturing packets
- Set what kind of packets Wireshark collects
- Syntax is **BPF-style** (Berkeley Packet Filter)
- Examples:
  - tcp port 80 → capture only HTTP traffic
  - host 192.168.1.1 → capture traffic to/from specific IP
  - net 10.0.0.0/8 → capture traffic from a subnet

- port 443 capture HTTPS

Set via: Capture Options > Capture Filter

Display Filters

- Applied **after** capturing packets
- Help **search, analyze, or isolate** specific flows
- Syntax is Wireshark-specific and more powerful than capture filters
- Examples:

- ip.addr == 192.168.1.5 → any traffic involving that IP
- tcp.flags.syn == 1 → TCP SYN packets
- http.request.method == "GET" → all HTTP GET requests
- dns.qry.name == "www.example.com" → DNS queries for a domain

📌 **Use in**: Filter bar or Find Packet dialog

🔍 2. Real-World Packet Capture Analysis in Wireshark

Let’s break down how Wireshark is used in actual traffic investigation:

HTTP Packet Analysis

- Apply filter: http
- Look for:
  - http.request.uri
  - http.host
  - http.response.code

You can reconstruct full HTTP conversations or extract files (e.g., images or downloads).

TCP Handshake

- Apply filter: tcp
- Look for 3-way handshake:
  - SYN
  - SYN-ACK
  - ACK

Identify retransmissions, zero-window, out-of-order packets.

DNS Packet Inspection

- Filter: dns
- Common fields:
  - dns.qry.name □ domain being queried
  - dns.flags.response □ if response or query
  - dns.a □ IP resolved

Use it to check for DNS hijacking or suspicious domains.

□ TLS/SSL Traffic

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Filter: ssl or tls

- You can analyze:
  - TLS versions used
  - Handshake details
  - Certificates
  - Weak ciphers (if visible)

Note: You **cannot see** decrypted content unless you have private keys or keylog file.

🔍 ICMP and Ping

- Filter: icmp
- Use to check connectivity issues

🔧 Tips for Efficient Filtering:

- Combine filters using logic:
  - and, or, not
  - ip.src == 10.0.0.1 and tcp.port == 22
- Bookmark frequently used filters

- Use color rules for protocol highlighting

🦈 Wireshark Use Cases

Use Case	Example
Network Troubleshooting	Packet loss, slow apps, latency
Security Monitoring	Suspicious DNS, port scanning, malware traffic
Protocol Learning	Analyze TCP handshakes, HTTP headers
Incident Response	Timeline reconstruction from PCAP
Penetration Testing	Analyze payloads and session hijacking

🔎 Wireshark Interface Overview

Section	Description
Packet List	Shows summary of each captured packet
Packet Details	Layer-wise breakdown (Ethernet, IP, TCP, etc.)
Hex Dump	Raw binary content
Filter Bar	Type display filter here
Capture Options	Choose interface, set limits